

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs Library

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GF180MCU_OSU_SC_GP12T3V3__ADDF_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT			OUTPUT	
A	B	CI	CO	SUM
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__addf_1	117.60000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	A	B	CI	CO	SUM
gf180mcu_osu_sc_gp12t3v3__addf_1	0.01546	0.01488	0.01140	15.45427	15.49830

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__addf_1	0.00000	0.00428	0.00452

Delay Information

Delay(ns) to CO rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addf_1	A->CO (RR)	0.17675	2.60758	67.96260
	B->CO (RR)	0.19570	2.89946	72.00290
	CI->CO (RR)	0.17735	2.64091	66.05660

Delay(ns) to CO falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addf_1	A->CO (FF)	0.20753	3.20178	75.87250
	B->CO (FF)	0.20249	3.52825	80.44350
	CI->CO (FF)	0.17984	3.37094	76.43470

Delay(ns) to SUM rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addf_1	A->SUM (-R)	0.34969	3.35785	77.22690
	B->SUM (-R)	0.34287	3.78468	83.92300
	CI->SUM (-R)	0.28916	3.54064	79.14530

Delay(ns) to SUM falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addf_1	A->SUM (-F)	0.21456	3.76432	85.27720
	B->SUM (-F)	0.24771	3.59114	82.54350
	CI->SUM (-F)	0.25755	3.34665	79.28210

Power Information

Internal switching power(pJ) to CO rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__addf_1	A	0.02415	0.20191	3.32863
	A	0.06628	0.23861	3.36524
	B	0.03106	0.18167	3.00477
	B	0.07205	0.22304	3.04864
	CI	0.02481	0.16539	2.60811
	CI	0.05342	0.19403	2.63823

Internal switching power(pJ) to CO falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__addf_1	A	0.07755	0.24922	3.36938
	A	0.03539	0.21004	3.32971
	B	0.07104	0.22719	3.02916
	B	0.02906	0.18573	2.98802
	CI	0.06455	0.20958	2.65833
	CI	0.03126	0.17658	2.62494

Internal switching power(pJ) to SUM rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__addf_1	A	0.00446	0.25228	4.97817
	A	0.08887	0.33598	5.04754
	B	0.00946	0.28490	5.39195
	B	0.09097	0.36556	5.46718
	CI	0.01720	0.32589	6.10489
	CI	0.09431	0.40212	6.17698

Internal switching power(pJ) to SUM falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__addf_1	A	0.09646	0.35092	5.05153
	A	0.00935	0.26465	4.97638
	B	0.09065	0.36678	5.46441
	B	0.01517	0.29229	5.40967
	CI	0.09525	0.40891	6.23517
	CI	0.02962	0.34378	6.17444

GF180MCU_OSU_SC_GP12T3V3__ADDH_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT	
A	B	CO	SUM
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__addh_1	67.62000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	A	B	CO	SUM
gf180mcu_osu_sc_gp12t3v3__addh_1	0.00768	0.00697	15.63292	15.55909

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__addh_1	0.00000	0.00342	0.00370

Delay Information

Delay(ns) to CO rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addh_1	A->CO (RR)	0.14242	2.65506	70.23780
	B->CO (RR)	0.14085	2.89956	73.94550

Delay(ns) to CO falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addh_1	A->CO (FF)	0.12607	2.99889	73.27020
	B->CO (FF)	0.11678	2.76518	68.80000

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addh_1	A->SUM (RR)	!B	0.14373	2.77934	72.13070
	A->SUM (FR)	B	0.20421	3.17137	76.09100
	B->SUM (RR)	!A	0.11815	2.42982	65.99510
	B->SUM (FR)	A	0.21849	2.92602	71.03440

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__addh_1	A->SUM (FF)	!B	0.14940	2.87220	71.51860
	A->SUM (RF)	B	0.21768	2.28375	56.90440
	B->SUM (FF)	!A	0.13471	3.16337	76.30660
	B->SUM (RF)	A	0.21588	2.59210	62.07060

Power Information

Internal switching power(pJ) to CO rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__addh_1	A	0.03098	0.21691	3.49101
	A	0.04914	0.23526	3.51037
	B	0.03562	0.20735	3.17373
	B	0.04753	0.21845	3.18584

Internal switching power(pJ) to CO falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__addh_1	A	0.05019	0.24125	3.53397
	A	0.03181	0.22299	3.51578
	B	0.04853	0.21813	3.17362
	B	0.03731	0.20717	3.16270

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__addh_1	A	B	0.05022	0.24181	3.53721
	A	B	0.03183	0.22361	3.51944
	A	!B	0.01369	0.30792	5.62210
	A	!B	0.06601	0.35998	5.66498
	B	A	0.04856	0.21794	3.17886
	B	A	0.03735	0.20697	3.16821
	B	!A	0.00832	0.26789	4.87956
	B	!A	0.04628	0.30523	4.91193

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__addh_1	A	B	0.03098	0.21657	3.48511
	A	B	0.04914	0.23481	3.50453
	A	!B	0.06275	0.35694	5.67214
	A	!B	0.01090	0.30583	5.62347
	B	A	0.03558	0.20830	3.16867
	B	A	0.04749	0.21948	3.18330
	B	!A	0.05449	0.31404	4.90909
	B	!A	0.01595	0.27582	4.87267

GF180MCU_OSU_SC_GP12T3V3__AND2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	0
1	0	0
1	1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__and2_1	35.28000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
gf180mcu_osu_sc_gp12t3v3__and2_1	0.00404	0.00403	15.46230

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__and2_1	0.00000	0.00144	0.00205

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__and2_1	A->Y (RR)	0.11156	2.76933	72.95150
	B->Y (RR)	0.11281	2.54895	69.30150

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__and2_1	A->Y (FF)	0.09591	2.63803	67.88340
	B->Y (FF)	0.10569	2.88975	72.38080

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__and2_1	A	0.01757	0.33774	5.93613
	A	0.04075	0.36081	5.95054
	B	0.01672	0.36605	6.57348
	B	0.04505	0.39398	6.59631

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__and2_1	A	0.03824	0.35876	5.95264
	A	0.01498	0.33635	5.93904
	B	0.04657	0.40111	6.59413
	B	0.01837	0.37416	6.56596

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__and2_1	(!B * !Y)	-0.01399	-0.01400	-0.01410
	(!B * !Y)	0.00180	0.00181	0.00173

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__and2_1	(!B * !Y)	0.01426	0.01427	0.01420
	(!B * !Y)	-0.00170	-0.00168	-0.00167

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__and2_1	(!A * !Y)	-0.01353	-0.01359	-0.01348
	(!A * !Y)	0.00651	0.00656	0.00642

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__and2_1	(!A * !Y)	0.01361	0.01388	0.01357
	(!A * !Y)	-0.00630	-0.00652	-0.00642

GF180MCU_OSU_SC_GP12T3V3__AOI21_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT			OUTPUT
A0	A1	B	Y
0	x	0	1
x	x	1	0
1	0	0	1
1	1	x	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__aoi21_1	35.28000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A0	A1	B	Y
gf180mcu_osu_sc_gp12t3v3__aoi21_1	0.00396	0.00398	0.00405	7.79615

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__aoi21_1	0.00000	0.00094	0.00180

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	A0->Y (FR)	0.11358	4.06061	85.09710
	A1->Y (FR)	0.09590	4.01155	84.48700
	B->Y (FR)	0.09280	4.75925	98.03850

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	A0->Y (RF)	0.08803	2.83721	60.56870
	A1->Y (RF)	0.08601	3.46547	72.46930
	B->Y (RF)	0.04535	2.50131	53.23420

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	A0	0.04068	0.18944	2.52993
	A0	0.00281	0.15140	2.48938
	A1	0.03193	0.16954	2.32616
	A1	-0.00101	0.13667	2.29211
	B	0.02197	0.19477	2.79248
	B	-0.00054	0.17199	2.76960

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	A0	0.00437	0.14213	2.28175
	A0	0.04223	0.18003	2.31956
	A1	0.00482	0.13118	2.00645
	A1	0.03754	0.16454	2.06764
	B	-0.00368	0.15605	2.54048
	B	0.01879	0.17950	2.60354

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	(A1 * B * !Y)	-0.01274	-0.01338	-0.01322
	(A1 * B * !Y)	0.00689	0.00655	0.00652
	(!A1 * B * !Y)	-0.01345	-0.01363	-0.01351
	(!A1 * B * !Y)	0.00642	0.00658	0.00645
	(!A1 * !B * Y)	-0.01349	-0.01356	-0.01351
	(!A1 * !B * Y)	0.00645	0.00649	0.00645

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	(A1 * B * !Y)	0.01333	0.01338	0.01322
	(A1 * B * !Y)	-0.00648	-0.00648	-0.00649
	(!A1 * B * !Y)	0.01355	0.01388	0.01357
	(!A1 * B * !Y)	-0.00637	-0.00652	-0.00645
	(!A1 * !B * Y)	0.01379	0.01388	0.01357
	(!A1 * !B * Y)	-0.00626	-0.00649	-0.00645

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	(B * !Y)	-0.01275	-0.01340	-0.01325
	(B * !Y)	0.00686	0.00657	0.00652
	(!A0 * !B * Y)	-0.01395	-0.01400	-0.01410
	(!A0 * !B * Y)	0.00180	0.00181	0.00173

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	(B * !Y)	0.01323	0.01340	0.01325
	(B * !Y)	-0.00648	-0.00648	-0.00649
	(!A0 * !B * Y)	0.01426	0.01427	0.01420
	(!A0 * !B * Y)	-0.00168	-0.00168	-0.00167

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	(A0 * A1 * !Y)	-0.00436	-0.00435	-0.00434
	(A0 * A1 * !Y)	0.00781	0.00783	0.00781

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi21_1	(A0 * A1 * !Y)	0.00477	0.00480	0.00453
	(A0 * A1 * !Y)	-0.00720	-0.00723	-0.00761

GF180MCU_OSU_SC_GP12T3V3__AOI22_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT				OUTPUT
A0	A1	B0	B1	Y
0	x	0	x	1
0	x	1	0	1
x	x	1	1	0
1	0	0	x	1
1	0	1	0	1
1	1	x	x	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__aoi22_1	44.10000

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A0	A1	B0	B1	Y
gf180mcu_osu_sc_gp12t3v3__aoi22_1	0.00396	0.00398	0.00405	0.00402	7.82120

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__aoi22_1	0.00000	0.00121	0.00180

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	A0->Y (FR)	0.14499	4.09314	85.32590
	A1->Y (FR)	0.12770	4.03930	84.71740
	B0->Y (FR)	0.10060	4.68168	96.67700
	B1->Y (FR)	0.11686	4.71977	97.06210

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	A0->Y (RF)	0.11868	2.90501	61.27190
	A1->Y (RF)	0.11653	3.53468	73.14220
	B0->Y (RF)	0.06740	3.45508	72.81130
	B1->Y (RF)	0.06860	2.82513	60.83520

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	A0	0.05049	0.20046	2.57131
	A0	0.00280	0.15277	2.52363
	A1	0.04165	0.18013	2.35999
	A1	-0.00101	0.13703	2.31734
	B0	0.02377	0.15940	2.17919
	B0	0.00004	0.13509	2.15130
	B1	0.03195	0.18038	2.34919
	B1	0.00330	0.15125	2.31739

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	A0	0.01349	0.16176	2.49819
	A0	0.06109	0.20933	2.54571
	A1	0.01385	0.15133	2.20708
	A1	0.05634	0.19387	2.25172
	B0	-0.00078	0.13185	2.08328
	B0	0.02296	0.15567	2.10811
	B1	-0.00157	0.14054	2.33130
	B1	0.02724	0.16943	2.36014

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(A1 * B0 * B1 * !Y)	-0.01261	-0.01333	-0.01320
	(A1 * B0 * B1 * !Y)	0.00684	0.00655	0.00652
	(!A1 * B0 * B1 * !Y)	-0.01352	-0.01358	-0.01348
	(!A1 * B0 * B1 * !Y)	0.00653	0.00656	0.00642
	(!A1 * B0 * !B1 * Y)	-0.01353	-0.01354	-0.01348
	(!A1 * B0 * !B1 * Y)	0.00653	0.00648	0.00642
	(!A1 * !B0 * Y)	-0.01353	-0.01354	-0.01348
	(!A1 * !B0 * Y)	0.00653	0.00648	0.00642

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(A1 * B0 * B1 * !Y)	0.01331	0.01333	0.01320
	(A1 * B0 * B1 * !Y)	-0.00649	-0.00648	-0.00649
	(!A1 * B0 * B1 * !Y)	0.01353	0.01388	0.01357
	(!A1 * B0 * B1 * !Y)	-0.00637	-0.00652	-0.00642
	(!A1 * B0 * !B1 * Y)	0.01379	0.01388	0.01357
	(!A1 * B0 * !B1 * Y)	-0.00625	-0.00648	-0.00642
	(!A1 * !B0 * Y)	0.01379	0.01388	0.01357
	(!A1 * !B0 * Y)	-0.00625	-0.00648	-0.00642

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(B0 * B1 * !Y)	-0.01262	-0.01332	-0.01317
	(B0 * B1 * !Y)	0.00679	0.00655	0.00652
	(!A0 * B0 * !B1 * Y)	-0.01394	-0.01399	-0.01410
	(!A0 * B0 * !B1 * Y)	0.00180	0.00181	0.00173
	(!A0 * !B0 * Y)	-0.01395	-0.01400	-0.01410
	(!A0 * !B0 * Y)	0.00180	0.00181	0.00173

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(B0 * B1 * !Y)	0.01329	0.01332	0.01317
	(B0 * B1 * !Y)	-0.00649	-0.00648	-0.00648
	(!A0 * B0 * !B1 * Y)	0.01424	0.01427	0.01420
	(!A0 * B0 * !B1 * Y)	-0.00167	-0.00168	-0.00167
	(!A0 * !B0 * Y)	0.01424	0.01427	0.01420
	(!A0 * !B0 * Y)	-0.00168	-0.00168	-0.00167

Passive power(pJ) for B0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(A0 * A1 * !Y)	-0.00434	-0.00434	-0.00434
	(A0 * A1 * !Y)	0.00777	0.00782	0.00781
	(!A1 * !B1 * Y)	-0.01408	-0.01408	-0.01410
	(!A1 * !B1 * Y)	0.00182	0.00183	0.00173
	(!A0 * A1 * !B1 * Y)	-0.01408	-0.01408	-0.01410
	(!A0 * A1 * !B1 * Y)	0.00183	0.00183	0.00173

Passive power(pJ) for B0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(A0 * A1 * !Y)	0.00493	0.00493	0.00458
	(A0 * A1 * !Y)	-0.00706	-0.00706	-0.00755
	(!A1 * !B1 * Y)	0.01412	0.01426	0.01420
	(!A1 * !B1 * Y)	-0.00166	-0.00168	-0.00167
	(!A0 * A1 * !B1 * Y)	0.01410	0.01426	0.01420
	(!A0 * A1 * !B1 * Y)	-0.00165	-0.00168	-0.00167

Passive power(pJ) for B1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(A0 * A1 * !Y)	-0.00437	-0.00435	-0.00434
	(A0 * A1 * !Y)	0.00783	0.00781	0.00781
	(!A1 * !B0 * Y)	-0.01354	-0.01353	-0.01352
	(!A1 * !B0 * Y)	0.00648	0.00650	0.00644
	(!A0 * A1 * !B0 * Y)	-0.01354	-0.01361	-0.01348
	(!A0 * A1 * !B0 * Y)	0.00652	0.00660	0.00643

Passive power(pJ) for B1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__aoi22_1	(A0 * A1 * !Y)	0.00491	0.00493	0.00458
	(A0 * A1 * !Y)	-0.00702	-0.00706	-0.00755
	(!A1 * !B0 * Y)	0.01359	0.01361	0.01357
	(!A1 * !B0 * Y)	-0.00633	-0.00649	-0.00644
	(!A0 * A1 * !B0 * Y)	0.01359	0.01361	0.01357
	(!A0 * A1 * !B0 * Y)	-0.00632	-0.00649	-0.00643

GF180MCU_OSU_SC_GP12T3V3__BUF_16

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__buf_16	135.24001

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__buf_16	0.00404	248.50584

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__buf_16	0.00000	0.01253	0.01497

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_16	A->Y (RR)	0.33318	1.75336	69.02080

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_16	A->Y (FF)	0.35881	2.44198	75.59810

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_16	A	0.71559	1.11342	7.51187
	A	0.73757	1.13326	7.53376

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_16	A	0.76949	1.10121	7.44253
	A	0.74743	1.07945	7.47469

GF180MCU_OSU_SC_GP12T3V3__BUF_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__buf_1	26.46000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__buf_1	0.00404	15.60333

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__buf_1	0.00000	0.00147	0.00147

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_1	A->Y (RR)	0.08202	2.37883	67.36110

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_1	A->Y (FF)	0.09009	2.90183	73.88930

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_1	A	0.01424	0.39279	6.93368
	A	0.03604	0.41375	6.95543

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_1	A	0.03631	0.41727	6.94970
	A	0.01450	0.39623	6.92784

GF180MCU_OSU_SC_GP12T3V3__BUF_2

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__buf_2	35.28000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__buf_2	0.00405	31.24748

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__buf_2	0.00000	0.00221	0.00237

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_2	A->Y (RR)	0.09368	1.91512	67.60510

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_2	A->Y (FF)	0.10236	2.49368	74.13340

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_2	A	0.03381	0.41361	6.95726
	A	0.05575	0.43402	6.97903

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_2	A	0.05399	0.43434	6.97355
	A	0.03200	0.41391	6.96379

GF180MCU_OSU_SC_GP12T3V3__BUF_4

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__buf_4	47.04000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__buf_4	0.00404	62.37794

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__buf_4	0.00000	0.00369	0.00417

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_4	A->Y (RR)	0.12436	1.63951	67.74940

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_4	A->Y (FF)	0.13517	2.26014	74.31160

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_4	A	0.08153	0.46440	7.00894
	A	0.10359	0.48540	7.03028

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_4	A	0.10065	0.48265	7.02004
	A	0.07851	0.46151	7.02242

GF180MCU_OSU_SC_GP12T3V3__BUF_8

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__buf_8	76.44000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__buf_8	0.00404	123.63758

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__buf_8	0.00000	0.00663	0.00777

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_8	A->Y (RR)	0.18962	1.56294	67.65870

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__buf_8	A->Y (FF)	0.20539	2.22077	74.29470

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_8	A	0.22424	0.62210	7.31047
	A	0.24621	0.64184	7.17867

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__buf_8	A	0.24577	0.62330	7.13522
	A	0.22378	0.60145	7.21107

GF180MCU_OSU_SC_GP12T3V3__CLKBUF_16

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	135.24001

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	0.00404	248.51486

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	0.00000	0.01253	0.01497

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	A->Y (RR)	0.33323	1.75350	69.02360

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	A->Y (FF)	0.35885	2.44211	75.60050

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	A	0.71569	1.11348	7.51196
	A	0.73767	1.13332	7.53383

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_16	A	0.76955	1.10124	7.44261
	A	0.74750	1.07949	7.47475

GF180MCU_OSU_SC_GP12T3V3__CLKBUF_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	26.46000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	0.00404	15.60333

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	0.00000	0.00147	0.00147

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	A->Y (RR)	0.08202	2.37883	67.36110

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	A->Y (FF)	0.09009	2.90183	73.88930

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	A	0.01424	0.39279	6.93368
	A	0.03604	0.41375	6.95543

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_1	A	0.03631	0.41727	6.94970
	A	0.01450	0.39623	6.92784

GF180MCU_OSU_SC_GP12T3V3__CLKBUF_2

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	35.28000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	0.00405	31.24748

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	0.00000	0.00221	0.00237

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	A->Y (RR)	0.09368	1.91512	67.60510

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	A->Y (FF)	0.10236	2.49368	74.13340

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	A	0.03381	0.41361	6.95726
	A	0.05575	0.43402	6.97903

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_2	A	0.05399	0.43434	6.97355
	A	0.03200	0.41391	6.96379

GF180MCU_OSU_SC_GP12T3V3__CLKBUF_4

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	47.04000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	0.00404	62.37794

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	0.00000	0.00369	0.00417

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	A->Y (RR)	0.12436	1.63951	67.74940

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	A->Y (FF)	0.13517	2.26014	74.31160

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	A	0.08153	0.46440	7.00894
	A	0.10359	0.48540	7.03028

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_4	A	0.10065	0.48265	7.02004
	A	0.07851	0.46151	7.02242

GF180MCU_OSU_SC_GP12T3V3__CLKBUF_8

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	76.44000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	0.00404	123.63758

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	0.00000	0.00663	0.00777

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	A->Y (RR)	0.18962	1.56294	67.65870

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	A->Y (FF)	0.20539	2.22077	74.29470

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	A	0.22424	0.62210	7.31047
	A	0.24621	0.64184	7.17867

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkbuf_8	A	0.24577	0.62330	7.13522
	A	0.22378	0.60145	7.21107

GF180MCU_OSU_SC_GP12T3V3__CLKINV_16

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkinv_16	126.42000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkinv_16	0.06677	234.89937

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkinv_16	0.00000	0.01179	0.01439

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_16	A->Y (FR)	0.03235	2.28876	98.41110

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_16	A->Y (RF)	0.02417	1.50661	83.67180

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_16	A	0.33521	4.38466	38.25270
	A	-0.02579	4.01862	37.86550

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_16	A	-0.06515	3.81519	34.45980
	A	0.29624	4.17840	34.82290

GF180MCU_OSU_SC_GP12T3V3__CLKINV_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkinv_1	20.58000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkinv_1	0.00404	14.68244

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkinv_1	0.00000	0.00074	0.00090

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_1	A->Y (FR)	0.04995	4.11040	98.41570

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_1	A->Y (RF)	0.04180	3.33251	83.67590

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_1	A	0.01985	0.20210	2.39002
	A	-0.00202	0.17833	2.36651

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_1	A	-0.00447	0.15442	2.15348
	A	0.01744	0.17776	2.17547

GF180MCU_OSU_SC_GP12T3V3__CLKINV_2

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkinv_2	26.46000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkinv_2	0.00808	29.36374

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkinv_2	0.00000	0.00147	0.00180

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_2	A->Y (FR)	0.04071	3.52273	98.41320

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_2	A->Y (RF)	0.03262	2.74856	83.67370

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_2	A	0.04005	0.44650	4.78011
	A	-0.00375	0.40085	4.73310

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_2	A	-0.00880	0.35376	4.30699
	A	0.03494	0.39879	4.35098

GF180MCU_OSU_SC_GP12T3V3__CLKINV_4

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkinv_4	41.16000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkinv_4	0.01616	58.72635

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkinv_4	0.00000	0.00295	0.00360

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_4	A->Y (FR)	0.03545	3.02967	98.41200

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_4	A->Y (RF)	0.02731	2.25833	83.67260

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_4	A	0.08045	0.94112	9.56029
	A	-0.00682	0.84864	9.46628

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_4	A	-0.01720	0.79012	8.61403
	A	0.07006	0.87965	8.70200

GF180MCU_OSU_SC_GP12T3V3__CLKINV_8

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__clkinv_8	70.56000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__clkinv_8	0.03236	117.45156

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__clkinv_8	0.00000	0.00590	0.00720

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_8	A->Y (FR)	0.03255	2.62820	98.41140

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__clkinv_8	A->Y (RF)	0.02439	1.84938	83.67200

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_8	A	0.16161	2.04091	19.12060
	A	-0.01294	1.86520	18.93270

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__clkinv_8	A	-0.03430	1.74534	17.22810
	A	0.14069	1.92329	17.40400

GF180MCU_OSU_SC_GP12T3V3__DFFN_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT	
D	CLK	Q	QN
0	F	0	1
1	F	1	0
x	x	IQ	IQN

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__dffn_1	120.54000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	D	CLK	Q	QN
gf180mcu_osu_sc_gp12t3v3__dffn_1	0.00395	0.00407	15.56757	15.73192

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__dffn_1	0.00000	100780.00000	174312.00000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK->Q (FR)	0.36894	6.31486	169.75700
	QN->Q (FR)	0.04995	4.19520	102.18800

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK->Q (FF)	0.65926	6.61154	169.08500
	QN->Q (RF)	0.04180	3.41465	87.05470

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK->QN (FR)	0.61931	3.52797	77.38850

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK->QN (FF)	0.32345	2.88944	67.87030

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	hold	CLK (F)	0.01742	0.10744	-15.97740
	setup	CLK (F)	0.31879	0.86130	29.09670

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	hold	CLK (F)	-0.31633	-1.82463	46.00980
	setup	CLK (F)	0.44205	1.82773	-31.51150

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	min_pulse_width	CLK ()	D	0.38414	4.44336	287.65300
	min_pulse_width	CLK ()	!D	0.36508	4.44336	287.26700

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	min_pulse_width	CLK ()	D	0.21733	4.44336	165.00100
	min_pulse_width	CLK ()	!D	0.14346	4.44336	165.00100

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK	0.09613	0.30540	4.57276
	CLK	0.08502	0.29437	4.56238

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK	0.14103	0.34266	4.67148
	CLK	0.13009	0.33080	4.64590

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK	0.14111	0.34147	4.66113
	CLK	0.13016	0.33048	4.63001

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	CLK	0.09601	0.30671	4.55022
	CLK	0.08490	0.29580	4.55149

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	$(CLK * Q * !QN) + (CLK * !Q * QN)$	0.15519	0.89381	12.51980
	$(CLK * Q * !QN) + (CLK * !Q * QN)$	0.17670	0.91535	12.54120
	$!CLK$	-0.01324	-0.01352	-0.01342
	$!CLK$	0.00666	0.00653	0.00646

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	$(CLK * Q * !QN) + (CLK * !Q * QN)$	0.22996	0.99699	12.90250
	$(CLK * Q * !QN) + (CLK * !Q * QN)$	0.20849	0.97544	12.88100
	$!CLK$	0.01360	0.01366	0.01343
	$!CLK$	-0.00637	-0.00648	-0.00645

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	$(D * Q * !QN)$	0.03617	0.43917	7.27886
	$(D * Q * !QN)$	0.05835	0.46144	7.30110
	$(D * !Q * QN)$	0.20005	0.57320	19.85630
	$(D * !Q * QN)$	0.22303	0.59636	19.87960
	$(!D * Q * !QN)$	0.23296	0.54296	7.53163
	$(!D * Q * !QN)$	0.25433	0.56435	7.55303
	$(!D * !Q * QN)$	0.04558	0.44926	7.28931
	$(!D * !Q * QN)$	0.06737	0.47116	7.31111

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffn_1	(D * Q * !QN)	0.06238	0.46702	7.29869
	(D * Q * !QN)	0.04029	0.44486	7.27624
	(!D * !Q * QN)	0.06740	0.47148	7.30729
	(!D * !Q * QN)	0.04553	0.44964	7.28546

GF180MCU_OSU_SC_GP12T3V3__DFFSR_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ees
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT				OUTPUT	
D	R	S	CLK	Q	QN
0	1	1	R	0	1
1	1	1	R	1	0
x	0	x	x	0	1
x	1	0	x	1	0
x	1	1	x	IQ	IQN

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__dffsr_1	158.75999

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)	
	D	R	S	CLK	Q	QN
gf180mcu_osu_sc_gp12t3v3__dffsr_1	0.00356	0.00407	0.00831	0.01155	15.67782	15.73825

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__dffsr_1	0.00000	0.00694	0.00849

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK->Q (RR)	0.34252	5.69307	157.35500
	QN->Q (FR)	0.04994	4.20522	102.66000
	R->Q (RR)	0.27655	5.72854	160.40400
	S->Q (FR)	0.25298	6.21376	168.13900

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK->Q (RF)	0.38729	5.67509	156.65000
	QN->Q (RF)	0.04180	3.42469	87.48170
	R->Q (FF)	0.23545	6.29954	169.98000

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK->QN (RR)	0.34547	2.56892	63.85290
	R->QN (FR)	0.19403	3.19788	77.17290

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK->QN (RF)	0.29407	2.24995	54.34650
	R->QN (RF)	0.22961	2.29220	57.44750
	S->QN (FF)	0.20489	2.76675	65.11800

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	hold	CLK (R)	-0.16201	-0.22885	2.10977
	setup	CLK (R)	0.17231	0.27749	-2.09677

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	hold	CLK (R)	-0.18491	-1.32982	-44.38470
	setup	CLK (R)	0.18857	1.33291	44.38840

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	hold	CLK (R)	(R * S)	-0.16201	-0.22885	2.10977
	setup	CLK (R)	(R * S)	0.17231	0.27749	-2.09677

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	hold	CLK (R)	(R * S)	-0.18491	-1.32982	-44.38470
	setup	CLK (R)	(R * S)	0.18857	1.33291	44.38840

Constraints(ns) for R rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	recovery	CLK (R)	0.05205	0.23664	0.59593
	removal	CLK (R)	-0.02801	-0.05567	-0.58889
	hold	S (R)	-0.19297	-0.57213	0.08211
	setup	S (R)	0.23335	0.72079	1.15006

Constraints(ns) for R rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	recovery	CLK (R)	(D * S)	0.05205	0.23664	0.59593
	removal	CLK (R)	(D * S)	-0.02801	-0.05567	-0.58889
	hold	S (R)	CLK	-0.19297	-0.57213	0.08211
	hold	S (R)	!CLK	-0.19581	-0.58450	-0.22910
	setup	S (R)	CLK	0.21389	0.69663	0.63468
	setup	S (R)	!CLK	0.23335	0.72079	1.15006

Constraints(ns) for R falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	min_pulse_width	R ()	(CLK * S)	0.14346	4.44336	165.00100
	min_pulse_width	R ()	(!CLK * S)	0.13631	4.44336	165.00100

Constraints(ns) for S rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	recovery	CLK (R)	0.04128	0.28223	5.82760
	removal	CLK (R)	-0.03805	-0.21648	-5.83471

Constraints(ns) for S rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	recovery	CLK (R)	(!D * R)	0.04128	0.28223	5.82760
	removal	CLK (R)	(!D * R)	-0.03805	-0.21648	-5.83471

Constraints(ns) for S falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	min_pulse_width	S ()	(CLK * R)	0.18874	4.44336	165.00100
	min_pulse_width	S ()	(!CLK * R)	0.20780	4.44336	165.00100

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	min_pulse_width	CLK ()	(D * R * S)	0.15299	4.44336	165.00100
	min_pulse_width	CLK ()	(!D * R * S)	0.16014	4.44336	165.00100

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	min_pulse_width	CLK ()	(D * R * S)	0.22686	4.44336	165.00100
	min_pulse_width	CLK ()	(!D * R * S)	0.16491	4.44336	165.00100

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK	0.04688	0.35288	5.55587
	CLK	0.07160	0.38422	5.59273
	R	0.06754	0.30221	4.70304
	R	0.09924	0.33282	4.75503
	S	0.07968	0.35792	5.10450
	S	0.06336	0.34205	5.08822

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK	0.05057	0.27147	4.52387
	CLK	0.07508	0.29534	4.54822
	R	0.10039	0.34384	4.85169
	R	0.08358	0.32709	4.82412

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK	0.05053	0.27081	4.50719
	CLK	0.07504	0.29491	4.53460
	R	0.10038	0.34367	4.83769
	R	0.08357	0.32604	4.80068

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK	0.04678	0.35421	5.52593
	CLK	0.07149	0.37915	5.55773
	R	0.08235	0.31541	4.69229
	R	0.09912	0.33226	4.70903
	S	0.07958	0.35795	5.07347
	S	0.06326	0.34170	5.08154

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK	-0.01322	-0.01344	-0.01329
	CLK	0.00463	0.00463	0.00461
	(!CLK * R * S * Q * !QN) + (!CLK * R * S * !Q * QN)	0.06355	0.35848	5.69610
	(!CLK * R * S * Q * !QN) + (!CLK * R * S * !Q * QN)	0.08718	0.38210	5.71946
	(!CLK * R * !S * Q * !QN)	0.02341	0.29505	5.26996
	(!CLK * R * !S * Q * !QN)	0.05321	0.32483	5.29961
	(!CLK * !R * S * !Q * QN)	0.02317	0.29499	5.27033
	(!CLK * !R * S * !Q * QN)	0.05300	0.32500	5.30018
	(!CLK * !R * !S * !Q * QN)	0.02337	0.29506	5.26997
	(!CLK * !R * !S * !Q * QN)	0.05319	0.32483	5.29961

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	CLK	0.01346	0.01352	0.01329
	CLK	-0.00445	-0.00458	-0.00455
	$(\text{!CLK} * \text{R} * \text{S} * \text{Q} * \text{!QN}) + (\text{!CLK} * \text{R} * \text{S} * \text{!Q} * \text{QN})$	0.08497	0.38783	5.75601
	$(\text{!CLK} * \text{R} * \text{S} * \text{Q} * \text{!QN}) + (\text{!CLK} * \text{R} * \text{S} * \text{!Q} * \text{QN})$	0.06148	0.36437	5.73264
	$(\text{!CLK} * \text{R} * \text{!S} * \text{Q} * \text{!QN})$	0.03876	0.31213	5.28734
	$(\text{!CLK} * \text{R} * \text{!S} * \text{Q} * \text{!QN})$	0.00909	0.28241	5.25786
	$(\text{!CLK} * \text{!R} * \text{S} * \text{!Q} * \text{QN})$	0.03905	0.31127	5.28778
	$(\text{!CLK} * \text{!R} * \text{S} * \text{!Q} * \text{QN})$	0.00929	0.28152	5.25817
	$(\text{!CLK} * \text{!R} * \text{!S} * \text{!Q} * \text{QN})$	0.03875	0.31213	5.28786
	$(\text{!CLK} * \text{!R} * \text{!S} * \text{!Q} * \text{QN})$	0.00908	0.28241	5.25809

Passive power(pJ) for R rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	$(\text{CLK} * \text{S} * \text{!Q} * \text{QN}) + (\text{!CLK} * \text{!D} * \text{S} * \text{!Q} * \text{QN})$	0.00538	0.37797	6.86143
	$(\text{CLK} * \text{S} * \text{!Q} * \text{QN}) + (\text{!CLK} * \text{!D} * \text{S} * \text{!Q} * \text{QN})$	0.02776	0.40032	6.88369
	$(\text{!CLK} * \text{D} * \text{S} * \text{!Q} * \text{QN})$	0.04427	0.43430	7.19873
	$(\text{!CLK} * \text{D} * \text{S} * \text{!Q} * \text{QN})$	0.06114	0.45110	7.21577

Passive power(pJ) for R falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	$(CLK * S * !Q * QN) + (!CLK * !D * S * !Q * QN)$	0.03562	0.41324	6.89545
	$(CLK * S * !Q * QN) + (!CLK * !D * S * !Q * QN)$	0.01344	0.39092	6.87329
	$(!CLK * D * S * !Q * QN)$	0.07318	0.46988	7.22421
	$(!CLK * D * S * !Q * QN)$	0.05625	0.45285	7.20680

Passive power(pJ) for S rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	$(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$	-0.02787	-0.02797	-0.02819
	$(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$	0.00383	0.00385	0.00366
	$(!R * !Q * QN)$	-0.02608	-0.02711	-0.02697
	$(!R * !Q * QN)$	0.01371	0.01316	0.01305
	$(!CLK * !D * R * Q * !QN)$	0.01818	0.30882	5.57959
	$(!CLK * !D * R * Q * !QN)$	0.05578	0.34660	5.61712

Passive power(pJ) for S falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	$(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$	0.02858	0.02855	0.02841
	$(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$	-0.00354	-0.00352	-0.00349
	$(!R * !Q * QN)$	0.02682	0.02726	0.02697
	$(!R * !Q * QN)$	-0.01291	-0.01294	-0.01296
	$(!CLK * !D * R * Q * !QN)$	0.05863	0.34576	5.61922
	$(!CLK * !D * R * Q * !QN)$	0.02095	0.30788	5.58160

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	$(D * R * Q * !QN)$	-0.00391	0.37003	6.85435
	$(D * R * Q * !QN)$	0.04266	0.41648	6.90085
	$(D * !R * S * !Q * QN)$	0.02405	0.40667	7.11611
	$(D * !R * S * !Q * QN)$	0.06805	0.44908	7.15708
	$(D * !R * !S * !Q * QN)$	0.02377	0.40654	7.11686
	$(D * !R * !S * !Q * QN)$	0.06785	0.44889	7.15667
	$(!D * R * S * !Q * QN) + (!D * !R * !Q * QN)$	-0.00505	0.36528	6.85076
	$(!D * R * S * !Q * QN) + (!D * !R * !Q * QN)$	0.04882	0.41919	6.90453
	$(!D * R * !S * Q * !QN)$	0.01386	0.64318	11.70040
	$(!D * R * !S * Q * !QN)$	0.07016	0.69991	11.75680

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dffsr_1	(D * R * S * !Q * QN)	0.12451	0.57493	9.66702
	(D * R * S * !Q * QN)	0.07704	0.52634	9.59693
	(D * R * Q * !QN)	0.04109	0.41706	6.90247
	(D * R * Q * !QN)	-0.00522	0.37049	6.85593
	(D * !R * S * !Q * QN)	0.08064	0.46316	7.15996
	(D * !R * S * !Q * QN)	0.03667	0.41999	7.11500
	(D * !R * !S * !Q * QN)	0.08095	0.46307	7.15722
	(D * !R * !S * !Q * QN)	0.03696	0.41968	7.11333
	(!D * R * S * Q * !QN)	0.11006	0.70602	9.83628
	(!D * R * S * Q * !QN)	0.05970	0.65536	9.78469
	(!D * R * S * !Q * QN) + (!D * !R * !Q * QN)	0.04998	0.42704	6.90589
	(!D * R * S * !Q * QN) + (!D * !R * !Q * QN)	-0.00371	0.37284	6.85195
	(!D * R * !S * Q * !QN)	0.05948	0.70159	11.75010
	(!D * R * !S * Q * !QN)	0.00312	0.64502	11.69380

GF180MCU_OSU_SC_GP12T3V3__DFF_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT	
D	CLK	Q	QN
0	R	0	1
1	R	1	0
x	x	IQ	IQN

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__dff_1	111.72000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	D	CLK	Q	QN
gf180mcu_osu_sc_gp12t3v3__dff_1	0.00394	0.01203	15.73103	15.73823

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__dff_1	0.00000	100780.00000	174312.00000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK->Q (RR)	0.29232	3.90849	118.43500
	QN->Q (FR)	0.04995	4.21004	102.88800

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK->Q (RF)	0.57905	5.62086	156.99700
	QN->Q (RF)	0.04180	3.42952	87.68530

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK->QN (RR)	0.53938	2.50387	63.63520

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK->QN (RF)	0.24685	0.44704	14.78600

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	hold	CLK (R)	-0.07398	-0.51182	-11.42710
	setup	CLK (R)	0.37732	2.02134	61.18360

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	hold	CLK (R)	-0.39742	1.00510	81.80850
	setup	CLK (R)	0.51535	3.52083	122.22500

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	min_pulse_width	CLK ()	D	0.19827	4.44336	165.00100
	min_pulse_width	CLK ()	!D	0.11963	4.44336	165.00100

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	min_pulse_width	CLK ()	D	0.39606	4.44336	288.85600
	min_pulse_width	CLK ()	!D	0.35793	4.51077	284.04300

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK	0.06160	0.26662	4.47452
	CLK	0.08451	0.28933	4.49723

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK	0.10656	0.27673	4.52267
	CLK	0.13213	0.30204	4.54822

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK	0.10670	0.27723	4.51837
	CLK	0.13226	0.30244	4.53455

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK	0.06152	0.26568	4.46849
	CLK	0.08443	0.28845	4.49534

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK	-0.01306	-0.01349	-0.01329
	CLK	0.00666	0.00652	0.00646
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.15152	0.89124	12.51460
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.19234	0.93196	12.55560

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	CLK	0.01344	0.01350	0.01329
	CLK	-0.00637	-0.00648	-0.00645
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.22862	0.99562	12.88440
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.18807	0.95487	12.84680

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	(D * Q * !QN)	-0.00220	0.36523	6.85299
	(D * Q * !QN)	0.04265	0.40983	6.89739
	(!D * !Q * QN)	-0.00489	0.36490	6.85053
	(!D * !Q * QN)	0.04799	0.41823	6.90365

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dff_1	(D * Q * !QN)	0.04190	0.41776	6.90383
	(D * Q * !QN)	-0.00255	0.37348	6.85911
	(D * !Q * QN)	0.20348	0.60920	7.35469
	(D * !Q * QN)	0.17158	0.58046	7.31782
	(!D * Q * !QN)	0.22913	0.66228	7.53008
	(!D * Q * !QN)	0.16539	0.59723	7.46490
	(!D * !Q * QN)	0.05108	0.41805	6.90460
	(!D * !Q * QN)	-0.00204	0.36472	6.85169

GF180MCU_OSU_SC_GP12T3V3__DLATN_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	Q
0	0	0
x	1	IQ
1	0	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__dlatn_1	91.14000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
gf180mcu_osu_sc_gp12t3v3__dlatn_1	0.00395	0.00405	15.49329

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__dlatn_1	0.00000	0.00481	0.00529

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	CLK->Q (FR)	0.29852	3.28778	76.23070
	D->Q (RR)	0.25448	2.45310	62.41010

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	CLK->Q (FF)	0.34973	3.03966	68.97380
	D->Q (FF)	0.28086	3.02289	69.80940

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	hold	CLK (R)	-0.08296	-0.25050	-4.00698
	setup	CLK (R)	0.09552	0.29689	7.01096

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	hold	CLK (R)	-0.07492	-0.38967	-6.16245
	setup	CLK (R)	0.08574	0.39276	6.16980

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	min_pulse_width	CLK ()	D	0.14346	4.44336	165.00100
	min_pulse_width	CLK ()	!D	0.16014	4.44336	165.00100

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	CLK	0.13043	0.56595	7.68050
	CLK	0.10955	0.54437	7.65199
	D	0.07499	0.44004	6.81991
	D	0.09648	0.46061	6.80578

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	CLK	0.13634	0.54231	7.36956
	CLK	0.11396	0.52073	7.35564
	D	0.10364	0.46972	6.79828
	D	0.08209	0.44880	6.77862

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	CLK	-0.01282	-0.01360	-0.01343
	CLK	0.00705	0.00663	0.00649

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	CLK	0.01354	0.01368	0.01343
	CLK	-0.00642	-0.00648	-0.00644

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	(D * Q)	0.02710	0.43291	7.27341
	(D * Q)	0.04889	0.45468	7.29517
	(!D * !Q)	0.03009	0.43663	7.27672
	(!D * !Q)	0.05213	0.45878	7.29872

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlatn_1	(D * Q)	0.04884	0.45680	7.28847
	(D * Q)	0.02713	0.43511	7.26683
	(!D * !Q)	0.05257	0.45977	7.29354
	(!D * !Q)	0.03046	0.43754	7.27144

GF180MCU_OSU_SC_GP12T3V3__DLAT_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	Q
x	0	IQ
0	1	0
1	1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__dlat_1	76.44000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
gf180mcu_osu_sc_gp12t3v3__dlat_1	0.00395	0.00897	15.71026

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__dlat_1	0.00000	0.00412	0.00471

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dlat_1	CLK->Q (RR)	0.22628	2.43748	61.37890
	D->Q (RR)	0.25330	2.47231	63.36240

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__dlat_1	CLK->Q (RF)	0.28915	2.21775	56.69900
	D->Q (FF)	0.28105	3.03993	70.66860

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	hold	CLK (F)	-0.14171	-0.69893	-20.45780
	setup	CLK (F)	0.14972	1.29407	44.32210

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	hold	CLK (F)	-0.12474	-0.14535	6.94599
	setup	CLK (F)	0.13513	0.15154	-6.94825

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	min_pulse_width	CLK ()	D	0.12201	4.44336	165.00100
	min_pulse_width	CLK ()	!D	0.14822	4.44336	165.00100

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	CLK	0.06732	0.65001	9.50834
	CLK	0.11199	0.69454	9.52895
	D	0.06874	0.43232	6.80558
	D	0.09653	0.45948	6.81192

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	CLK	0.09021	0.47434	7.20560
	CLK	0.11672	0.50120	7.23500
	D	0.11037	0.47618	6.80523
	D	0.08205	0.44867	6.77858

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	!CLK	-0.01282	-0.01360	-0.01343
	!CLK	0.00701	0.00659	0.00644

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	!CLK	0.01354	0.01368	0.01343
	!CLK	-0.00636	-0.00649	-0.00642

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	(D * Q)	-0.00447	0.37267	6.85868
	(D * Q)	0.03003	0.40728	6.89326
	(!D * !Q)	-0.00447	0.36967	6.85764
	(!D * !Q)	0.03324	0.40767	6.89543

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__dlat_1	(D * Q)	0.03285	0.41629	6.89759
	(D * Q)	-0.00156	0.38186	6.86298
	(!D * !Q)	0.03562	0.41608	6.89811
	(!D * !Q)	-0.00218	0.37817	6.86018

GF180MCU_OSU_SC_GP12T3V3__INV_16

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__inv_16	126.42000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__inv_16	0.06677	234.89937

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__inv_16	0.00000	0.01179	0.01439

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_16	A->Y (FR)	0.03235	2.28876	98.41110

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_16	A->Y (RF)	0.02417	1.50661	83.67180

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_16	A	0.33521	4.38466	38.25270
	A	-0.02579	4.01862	37.86550

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_16	A	-0.06515	3.81519	34.45980
	A	0.29624	4.17840	34.82290

GF180MCU_OSU_SC_GP12T3V3__INV_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__inv_1	20.58000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__inv_1	0.00404	14.68244

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__inv_1	0.00000	0.00074	0.00090

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_1	A->Y (FR)	0.04995	4.11040	98.41570

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_1	A->Y (RF)	0.04180	3.33251	83.67590

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_1	A	0.01985	0.20210	2.39002
	A	-0.00202	0.17833	2.36651

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_1	A	-0.00447	0.15442	2.15348
	A	0.01744	0.17776	2.17547

GF180MCU_OSU_SC_GP12T3V3__INV_2

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__inv_2	26.46000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__inv_2	0.00808	29.36374

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__inv_2	0.00000	0.00147	0.00180

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_2	A->Y (FR)	0.04071	3.52273	98.41320

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_2	A->Y (RF)	0.03262	2.74856	83.67370

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_2	A	0.04005	0.44650	4.78011
	A	-0.00375	0.40085	4.73310

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_2	A	-0.00880	0.35376	4.30699
	A	0.03494	0.39879	4.35098

GF180MCU_OSU_SC_GP12T3V3__INV_4

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__inv_4	41.16000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__inv_4	0.01616	58.72635

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__inv_4	0.00000	0.00295	0.00360

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_4	A->Y (FR)	0.03545	3.02967	98.41200

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_4	A->Y (RF)	0.02731	2.25833	83.67260

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_4	A	0.08045	0.94112	9.56029
	A	-0.00682	0.84864	9.46628

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_4	A	-0.01720	0.79012	8.61403
	A	0.07006	0.87965	8.70200

GF180MCU_OSU_SC_GP12T3V3__INV_8

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__inv_8	70.56000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
gf180mcu_osu_sc_gp12t3v3__inv_8	0.03236	117.45156

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__inv_8	0.00000	0.00590	0.00720

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_8	A->Y (FR)	0.03255	2.62820	98.41140

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__inv_8	A->Y (RF)	0.02439	1.84938	83.67200

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_8	A	0.16161	2.04091	19.12060
	A	-0.01294	1.86520	18.93270

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__inv_8	A	-0.03430	1.74534	17.22810
	A	0.14069	1.92329	17.40400

GF180MCU_OSU_SC_GP12T3V3__MUX2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT	OUTPUT
S	Y
x	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__mux2_1	41.16000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	S	Y
gf180mcu_osu_sc_gp12t3v3__mux2_1	0.00682	1.80000

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__mux2_1	0.00000	0.00095	0.00112

Passive Power Information

Passive power(pJ) for S rising :

Cell Name	Power(pJ)		
	first	mid	last
gf180mcu_osu_sc_gp12t3v3__mux2_1	-0.00500	0.37220	6.85820
	0.03196	0.40949	6.89531

Passive power(pJ) for S falling :

Cell Name	Power(pJ)		
	first	mid	last
gf180mcu_osu_sc_gp12t3v3__mux2_1	0.03412	0.41014	6.89776
	-0.00296	0.37301	6.86067

GF180MCU_OSU_SC_GP12T3V3__NAND2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	1
1	0	1
1	1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__nand2_1	26.46000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
gf180mcu_osu_sc_gp12t3v3__nand2_1	0.00404	0.00402	10.44023

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__nand2_1	0.00000	0.00078	0.00115

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__nand2_1	A->Y (FR)	0.05501	3.58983	79.20290
	B->Y (FR)	0.06349	3.62318	79.50170

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__nand2_1	A->Y (RF)	0.06305	3.94719	89.80080
	B->Y (RF)	0.06392	3.30633	78.16810

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nand2_1	A	0.02148	0.16638	2.19715
	A	-0.00162	0.14316	2.15837
	B	0.02981	0.18716	2.38040
	B	0.00159	0.15870	2.33677

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nand2_1	A	-0.00158	0.14274	2.10089
	A	0.02157	0.16619	2.12769
	B	-0.00247	0.15317	2.37665
	B	0.02577	0.18166	2.41500

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nand2_1	(!B * Y)	-0.01402	-0.01400	-0.01410
	(!B * Y)	0.00181	0.00181	0.00172

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nand2_1	(!B * Y)	0.01418	0.01426	0.01420
	(!B * Y)	-0.00167	-0.00168	-0.00167

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nand2_1	(!A * Y)	-0.01349	-0.01357	-0.01353
	(!A * Y)	0.00646	0.00656	0.00648

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nand2_1	(!A * Y)	0.01362	0.01388	0.01357
	(!A * Y)	-0.00631	-0.00652	-0.00646

GF180MCU_OSU_SC_GP12T3V3__NOR2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
x	1	0
1	x	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__nor2_1	23.52000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
gf180mcu_osu_sc_gp12t3v3__nor2_1	0.00399	0.00404	7.76937

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__nor2_1	0.00000	0.00082	0.00180

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__nor2_1	A->Y (FR)	0.08956	4.11832	86.27420
	B->Y (FR)	0.07533	4.73728	97.81030

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__nor2_1	A->Y (RF)	0.05878	2.51256	53.01870
	B->Y (RF)	0.04637	2.45366	52.46390

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nor2_1	A	0.03038	0.20714	2.99726
	A	-0.00150	0.17520	2.96359
	B	0.02167	0.17319	2.46298
	B	-0.00084	0.15041	2.44022

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nor2_1	A	0.00238	0.15709	2.49191
	A	0.03425	0.18971	2.53682
	B	-0.00315	0.13480	2.20967
	B	0.01938	0.15800	2.26171

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nor2_1	(B * !Y)	-0.01267	-0.01342	-0.01329
	(B * !Y)	0.00701	0.00657	0.00652

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nor2_1	(B * !Y)	0.01339	0.01342	0.01329
	(B * !Y)	-0.00650	-0.00648	-0.00648

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nor2_1	(A * !Y)	-0.00435	-0.00436	-0.00434
	(A * !Y)	0.00780	0.00781	0.00781

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__nor2_1	(A * !Y)	0.00470	0.00468	0.00448
	(A * !Y)	-0.00745	-0.00741	-0.00767

GF180MCU_OSU_SC_GP12T3V3__OAI21_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT			OUTPUT
A0	A1	B	Y
0	0	x	1
x	1	0	1
x	1	1	0
1	x	0	1
1	x	1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__oai21_1	32.34000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A0	A1	B	Y
gf180mcu_osu_sc_gp12t3v3__oai21_1	0.00397	0.00402	0.00404	7.88413

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__oai21_1	0.00000	0.00096	0.00152

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__oai21_1	A0->Y (FR)	0.11515	4.08209	85.78900
	A1->Y (FR)	0.09950	4.71649	97.57670
	B->Y (FR)	0.05463	3.29978	67.76420

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__oai21_1	A0->Y (RF)	0.09054	2.85235	61.12010
	A1->Y (RF)	0.06964	2.78764	60.50820
	B->Y (RF)	0.08723	3.55150	74.10740

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	A0	0.04072	0.18973	2.51348
	A0	0.00252	0.15142	2.47536
	A1	0.03146	0.15884	2.06542
	A1	0.00274	0.13012	2.03669
	B	0.02125	0.19884	2.83692
	B	-0.00185	0.17567	2.80625

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	A0	0.00507	0.14241	2.26661
	A0	0.04321	0.18065	2.30432
	A1	-0.00114	0.12204	2.02228
	A1	0.02755	0.15088	2.05100
	B	-0.00133	0.16864	2.68954
	B	0.02185	0.19271	2.71307

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	(A1 * B * !Y)	-0.01271	-0.01352	-0.01329
	(A1 * B * !Y)	0.00714	0.00657	0.00652
	(A1 * !B * Y)	-0.01317	-0.01341	-0.01328
	(A1 * !B * Y)	0.00649	0.00657	0.00652
	(!A1 * !B * Y)	-0.01354	-0.01355	-0.01353
	(!A1 * !B * Y)	0.00650	0.00655	0.00646

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	(A1 * B * !Y)	0.01332	0.01353	0.01329
	(A1 * B * !Y)	-0.00648	-0.00648	-0.00647
	(A1 * !B * Y)	0.01329	0.01341	0.01328
	(A1 * !B * Y)	-0.00649	-0.00650	-0.00649
	(!A1 * !B * Y)	0.01360	0.01359	0.01357
	(!A1 * !B * Y)	-0.00633	-0.00648	-0.00646

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	(A0 * B * !Y)	-0.00435	-0.00435	-0.00434
	(A0 * B * !Y)	0.00781	0.00781	0.00781
	(!B * Y)	-0.01321	-0.01346	-0.01326
	(!B * Y)	0.00657	0.00661	0.00652

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	(A0 * B * !Y)	0.00465	0.00468	0.00448
	(A0 * B * !Y)	-0.00737	-0.00741	-0.00767
	(!B * Y)	0.01321	0.01352	0.01326
	(!B * Y)	-0.00648	-0.00649	-0.00649

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	(!A0 * !A1 * Y)	-0.01395	-0.01392	-0.01408
	(!A0 * !A1 * Y)	0.00187	0.00187	0.00174

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai21_1	(!A0 * !A1 * Y)	0.01421	0.01425	0.01420
	(!A0 * !A1 * Y)	-0.00168	-0.00168	-0.00167

GF180MCU_OSU_SC_GP12T3V3__OAI22_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT				OUTPUT
A0	A1	B0	B1	Y
0	0	x	x	1
x	1	0	0	1
x	1	x	1	0
x	1	1	x	0
1	x	0	0	1
1	x	x	1	0
1	x	1	x	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__oai22_1	44.10000

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A0	A1	B0	B1	Y
gf180mcu_osu_sc_gp12t3v3__oai22_1	0.00396	0.00403	0.00405	0.00399	7.84087

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__oai22_1	0.00000	0.00125	0.00180

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__oai22_1	A0->Y (FR)	0.13551	4.12730	86.10960
	A1->Y (FR)	0.11973	4.76150	97.77950
	B0->Y (FR)	0.08298	4.71081	97.51380
	B1->Y (FR)	0.09752	4.06839	85.71610

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__oai22_1	A0->Y (RF)	0.12307	2.88058	60.88040
	A1->Y (RF)	0.10096	2.81945	60.27070
	B0->Y (RF)	0.09323	3.44390	72.32760
	B1->Y (RF)	0.11441	3.50469	72.78850

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	A0	0.05054	0.20891	2.74572
	A0	0.00722	0.16561	2.70089
	A1	0.04128	0.17927	2.25632
	A1	0.00739	0.14502	2.22231
	B0	0.02320	0.15703	2.16357
	B0	-0.00053	0.13309	2.13974
	B1	0.03211	0.18460	2.60981
	B1	-0.00113	0.15138	2.57660

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	A0	0.00508	0.14417	2.30038
	A0	0.04834	0.18857	2.35606
	A1	-0.00114	0.12363	2.05125
	A1	0.03275	0.15808	2.09821
	B0	0.00007	0.12607	1.99193
	B0	0.02387	0.15030	2.02516
	B1	0.00572	0.14487	2.21341
	B1	0.03883	0.17817	2.25310

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A1 * B0 * !Y)	-0.01267	-0.01352	-0.01327
	(A1 * B0 * !Y)	0.00716	0.00657	0.00652
	(A1 * !B0 * B1 * !Y)	-0.01270	-0.01352	-0.01328
	(A1 * !B0 * B1 * !Y)	0.00715	0.00657	0.00652
	(A1 * !B0 * !B1 * Y)	-0.01327	-0.01341	-0.01328
	(A1 * !B0 * !B1 * Y)	0.00658	0.00657	0.00652
	(!A1 * !B0 * !B1 * Y)	-0.01353	-0.01358	-0.01351
	(!A1 * !B0 * !B1 * Y)	0.00648	0.00655	0.00643

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A1 * B0 * !Y)	0.01339	0.01353	0.01327
	(A1 * B0 * !Y)	-0.00649	-0.00648	-0.00647
	(A1 * !B0 * B1 * !Y)	0.01330	0.01353	0.01328
	(A1 * !B0 * B1 * !Y)	-0.00648	-0.00648	-0.00647
	(A1 * !B0 * !B1 * Y)	0.01338	0.01341	0.01328
	(A1 * !B0 * !B1 * Y)	-0.00651	-0.00649	-0.00649
	(!A1 * !B0 * !B1 * Y)	0.01369	0.01360	0.01357
	(!A1 * !B0 * !B1 * Y)	-0.00629	-0.00646	-0.00643

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A0 * B0 * !Y)	-0.00443	-0.00435	-0.00434
	(A0 * B0 * !Y)	0.00789	0.00781	0.00781
	(A0 * !B0 * B1 * !Y)	-0.00432	-0.00435	-0.00434
	(A0 * !B0 * B1 * !Y)	0.00777	0.00781	0.00781
	(!B0 * !B1 * Y)	-0.01317	-0.01344	-0.01317
	(!B0 * !B1 * Y)	0.00656	0.00652	0.00652

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A0 * B0 * !Y)	0.00469	0.00468	0.00448
	(A0 * B0 * !Y)	-0.00741	-0.00740	-0.00767
	(A0 * !B0 * B1 * !Y)	0.00465	0.00468	0.00448
	(A0 * !B0 * B1 * !Y)	-0.00738	-0.00740	-0.00767
	(!B0 * !B1 * Y)	0.01317	0.01344	0.01317
	(!B0 * !B1 * Y)	-0.00649	-0.00650	-0.00649

Passive power(pJ) for B0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A1 * B1 * !Y)	-0.00438	-0.00435	-0.00434
	(A1 * B1 * !Y)	0.00780	0.00781	0.00781
	(A0 * !A1 * B1 * !Y)	-0.00438	-0.00435	-0.00434
	(A0 * !A1 * B1 * !Y)	0.00780	0.00781	0.00781
	(!A0 * !A1 * Y)	-0.01389	-0.01411	-0.01399
	(!A0 * !A1 * Y)	0.00163	0.00164	0.00163

Passive power(pJ) for B0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A1 * B1 * !Y)	0.00467	0.00468	0.00448
	(A1 * B1 * !Y)	-0.00740	-0.00741	-0.00767
	(A0 * !A1 * B1 * !Y)	0.00464	0.00468	0.00449
	(A0 * !A1 * B1 * !Y)	-0.00738	-0.00740	-0.00766
	(!A0 * !A1 * Y)	0.01409	0.01430	0.01399
	(!A0 * !A1 * Y)	-0.00144	-0.00146	-0.00153

Passive power(pJ) for B1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A1 * B0 * !Y)	-0.01272	-0.01352	-0.01328
	(A1 * B0 * !Y)	0.00710	0.00657	0.00652
	(A0 * !A1 * B0 * !Y)	-0.01270	-0.01342	-0.01332
	(A0 * !A1 * B0 * !Y)	0.00711	0.00657	0.00652
	(!A0 * !A1 * Y)	-0.01388	-0.01423	-0.01406
	(!A0 * !A1 * Y)	0.00163	0.00166	0.00163

Passive power(pJ) for B1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai22_1	(A1 * B0 * !Y)	0.01338	0.01353	0.01328
	(A1 * B0 * !Y)	-0.00649	-0.00648	-0.00647
	(A0 * !A1 * B0 * !Y)	0.01335	0.01342	0.01332
	(A0 * !A1 * B0 * !Y)	-0.00649	-0.00649	-0.00648
	(!A0 * !A1 * Y)	0.01434	0.01433	0.01412
	(!A0 * !A1 * Y)	-0.00145	-0.00145	-0.00153

GF180MCU_OSU_SC_GP12T3V3__OAI31_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT				OUTPUT
A0	A1	A2	B	Y
0	0	0	x	1
0	x	1	0	1
0	x	1	1	0
x	1	x	0	1
x	1	x	1	0
1	x	x	0	1
1	x	x	1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__oai31_1	41.16000

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A0	A1	A2	B	Y
gf180mcu_osu_sc_gp12t3v3__oai31_1	0.00395	0.00395	0.00402	0.00404	5.43109

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__oai31_1	0.00000	0.00101	0.00216

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__oai31_1	A0->Y (FR)	0.19021	4.38352	82.47430
	A1->Y (FR)	0.17247	4.80028	90.16140
	A2->Y (FR)	0.13197	5.23611	98.60270
	B->Y (FR)	0.05446	2.93267	55.17650

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__oai31_1	A0->Y (RF)	0.10558	2.37030	44.27920
	A1->Y (RF)	0.09666	2.30627	43.54280
	A2->Y (RF)	0.07361	2.24996	42.98220
	B->Y (RF)	0.09976	3.09566	57.61430

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	A0	0.05227	0.20430	2.92659
	A0	0.00427	0.15595	2.87695
	A1	0.04289	0.16833	2.32376
	A1	0.00422	0.12969	2.28429
	A2	0.03354	0.14912	2.04358
	A2	0.00430	0.11980	2.01398
	B	0.02119	0.22263	3.46862
	B	-0.00194	0.19930	3.43092

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	A0	0.01254	0.14324	2.39546
	A0	0.06040	0.19133	2.45671
	A1	0.00663	0.12419	2.13252
	A1	0.04523	0.16304	2.18426
	A2	-0.00061	0.10817	1.92571
	A2	0.02866	0.13745	1.96762
	B	-0.00125	0.19084	3.29681
	B	0.02196	0.21485	3.33324

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	$(A1 * A2 * B * !Y)$	-0.01314	-0.01341	-0.01329
	$(A1 * A2 * B * !Y)$	0.00656	0.00658	0.00652
	$(A1 * !B * Y)$	-0.01318	-0.01352	-0.01326
	$(A1 * !B * Y)$	0.00652	0.00657	0.00652
	$(A1 * !A2 * B * !Y) + (!A1 * A2 * B * !Y)$	-0.01299	-0.01352	-0.01327
	$(A1 * !A2 * B * !Y) + (!A1 * A2 * B * !Y)$	0.00673	0.00657	0.00652
	$(!A1 * A2 * !B * Y)$	-0.01222	-0.01289	-0.01288
	$(!A1 * A2 * !B * Y)$	0.00654	0.00657	0.00652
	$(!A1 * !A2 * !B * Y)$	-0.01354	-0.01353	-0.01351
	$(!A1 * !A2 * !B * Y)$	0.00649	0.00651	0.00644

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	$(A1 * A2 * B * !Y)$	0.01325	0.01341	0.01329
	$(A1 * A2 * B * !Y)$	-0.00646	-0.00649	-0.00649
	$(A1 * !B * Y)$	0.01329	0.01352	0.01326
	$(A1 * !B * Y)$	-0.00650	-0.00649	-0.00649
	$(A1 * !A2 * B * !Y) + (!A1 * A2 * B * !Y)$	0.01329	0.01352	0.01327
	$(A1 * !A2 * B * !Y) + (!A1 * A2 * B * !Y)$	-0.00649	-0.00649	-0.00649
	$(!A1 * A2 * !B * Y)$	0.01291	0.01289	0.01288
	$(!A1 * A2 * !B * Y)$	-0.00648	-0.00648	-0.00649
	$(!A1 * !A2 * !B * Y)$	0.01368	0.01360	0.01357
	$(!A1 * !A2 * !B * Y)$	-0.00630	-0.00646	-0.00644

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	$(A2 * !B * Y)$	-0.00788	-0.00786	-0.00779
	$(A2 * !B * Y)$	0.00653	0.00656	0.00652
	$(A0 * B * !Y) + (!A0 * A2 * B * !Y)$	-0.00696	-0.00698	-0.00695
	$(A0 * B * !Y) + (!A0 * A2 * B * !Y)$	0.00735	0.00732	0.00686
	$(!A2 * !B * Y)$	-0.01324	-0.01346	-0.01319
	$(!A2 * !B * Y)$	0.00656	0.00653	0.00652

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	$(A2 * !B * Y)$	0.00788	0.00786	0.00779
	$(A2 * !B * Y)$	-0.00647	-0.00649	-0.00649
	$(A0 * B * !Y) + (!A0 * A2 * B * !Y)$	0.00696	0.00698	0.00695
	$(A0 * B * !Y) + (!A0 * A2 * B * !Y)$	-0.00649	-0.00652	-0.00649
	$(!A2 * !B * Y)$	0.01324	0.01346	0.01319
	$(!A2 * !B * Y)$	-0.00649	-0.00650	-0.00649

Passive power(pJ) for A2 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	(A1 * B * !Y)	-0.00433	-0.00435	-0.00434
	(A1 * B * !Y)	0.00778	0.00781	0.00781
	(A1 * !B * Y)	-0.01310	-0.01346	-0.01323
	(A1 * !B * Y)	0.00657	0.00661	0.00652
	(A0 * !A1 * B * !Y)	-0.00426	-0.00426	-0.00425
	(A0 * !A1 * B * !Y)	0.00781	0.00780	0.00780
	(!A1 * !B * Y)	-0.01186	-0.01268	-0.01256
	(!A1 * !B * Y)	0.00650	0.00658	0.00652

Passive power(pJ) for A2 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	(A1 * B * !Y)	0.00462	0.00468	0.00448
	(A1 * B * !Y)	-0.00735	-0.00741	-0.00767
	(A1 * !B * Y)	0.01329	0.01352	0.01323
	(A1 * !B * Y)	-0.00651	-0.00651	-0.00649
	(A0 * !A1 * B * !Y)	0.00472	0.00475	0.00435
	(A0 * !A1 * B * !Y)	-0.00667	-0.00673	-0.00739
	(!A1 * !B * Y)	0.01251	0.01268	0.01256
	(!A1 * !B * Y)	-0.00649	-0.00649	-0.00649

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	(!A0 * !A1 * !A2 * Y)	-0.01386	-0.01384	-0.01405
	(!A0 * !A1 * !A2 * Y)	0.00194	0.00193	0.00177

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__oai31_1	(!A0 * !A1 * !A2 * Y)	0.01421	0.01426	0.01420
	(!A0 * !A1 * !A2 * Y)	-0.00168	-0.00168	-0.00167

GF180MCU_OSU_SC_GP12T3V3__OR2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__or2_1	32.34000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
gf180mcu_osu_sc_gp12t3v3__or2_1	0.00404	0.00399	15.52995

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__or2_1	0.00000	0.00164	0.00237

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__or2_1	A->Y (RR)	0.08702	2.01471	59.55870
	B->Y (RR)	0.10192	2.34896	65.55750

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__or2_1	A->Y (FF)	0.12502	3.37857	81.20600
	B->Y (FF)	0.13946	3.12073	76.95490

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__or2_1	A	0.01553	0.30950	5.50672
	A	0.03802	0.33135	5.50706
	B	0.02130	0.37127	6.53283
	B	0.05327	0.40228	6.55744

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__or2_1	A	0.03932	0.33551	5.49120
	A	0.01672	0.31392	5.47204
	B	0.04810	0.39080	6.53810
	B	0.01607	0.35888	6.51616

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__or2_1	(B * Y)	-0.00435	-0.00435	-0.00434
	(B * Y)	0.00780	0.00781	0.00781

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__or2_1	(B * Y)	0.00471	0.00468	0.00448
	(B * Y)	-0.00742	-0.00740	-0.00767

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__or2_1	(A * Y)	-0.01271	-0.01352	-0.01328
	(A * Y)	0.00714	0.00657	0.00652

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__or2_1	(A * Y)	0.01340	0.01353	0.01328
	(A * Y)	-0.00649	-0.00648	-0.00647

GF180MCU_OSU_SC_GP12T3V3__TBUF_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	EN	Y
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__tbuf_1	44.10000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	EN	Y
gf180mcu_osu_sc_gp12t3v3__tbuf_1	0.00404	0.00535	8.22652

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__tbuf_1	0.00000	0.00183	0.00201

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	A->Y (RR)	0.14286	2.96671	64.48500
	EN->Y (FR)	0.06209	3.64239	65.65660
	EN->Y (RR)	0.08973	2.97846	66.82990

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	A->Y (FF)	0.13160	2.96992	60.81760
	EN->Y (FF)	0.07523	3.64239	65.65660
	EN->Y (RF)	0.04019	3.14846	70.40980

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	A	0.03307	0.40982	6.96116
	A	0.04972	0.42615	6.97209
	EN	0.01597	0.38706	6.91208
	EN	0.03909	0.41013	6.93339

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	A	0.04301	0.42219	6.93951
	A	0.02625	0.40550	6.93223
	EN	0.00865	0.38665	6.91171
	EN	0.03754	0.41676	6.94411

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	!EN	0.00900	0.38466	6.86939
	!EN	0.03097	0.40673	6.89135

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	!EN	0.02645	0.40435	6.88807
	!EN	0.00438	0.38225	6.86607

Passive power(pJ) for EN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	(A * Y)	0.00808	0.38657	6.87217
	(A * Y)	0.03233	0.41100	6.89660
	(!A * !Y)	0.00025	0.37777	6.86687
	(!A * !Y)	0.02866	0.40631	6.89527

Passive power(pJ) for EN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tbuf_1	(A * Y)	0.02084	0.40044	6.88427
	(A * Y)	-0.00356	0.37589	6.85986
	(!A * !Y)	0.02126	0.40545	6.88887
	(!A * !Y)	-0.00716	0.37696	6.86034

GF180MCU_OSU_SC_GP12T3V3__TIEH

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__tieh	20.58000

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
gf180mcu_osu_sc_gp12t3v3__tieh	34.42027

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__tieh	0.00000	0.00000	0.00000

GF180MCU_OSU_SC_GP12T3V3__TIEL

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__tiel	20.58000

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
gf180mcu_osu_sc_gp12t3v3__tiel	51.62853

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__tiel	0.00000	0.00000	0.00000

GF180MCU_OSU_SC_GP12T3V3__TINV_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	EN	Y
-	0	HiZ
0	1	1
1	1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__tinv_1	32.34000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	EN	Y
gf180mcu_osu_sc_gp12t3v3__tinv_1	0.00396	0.00536	8.01146

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__tinv_1	0.00000	0.00109	0.00144

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__tinv_1	A->Y (FR)	0.10541	4.10171	86.79460
	EN->Y (FR)	0.06209	3.64239	65.65660
	EN->Y (RR)	0.08984	2.93386	65.01830

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__tinv_1	A->Y (RF)	0.08078	2.86550	61.92080
	EN->Y (FF)	0.07514	3.64239	65.65660
	EN->Y (RF)	0.04021	3.10444	68.90440

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tinvt_1	A	0.03718	0.18707	2.49375
	A	0.01053	0.16026	2.46488
	EN	0.01592	0.38790	6.91353
	EN	0.03861	0.41027	6.93032

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tinvt_1	A	0.00170	0.13891	2.24365
	A	0.02829	0.16550	2.27567
	EN	0.00771	0.38578	6.90885
	EN	0.03754	0.41627	6.94765

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tinvt_1	!EN	-0.01330	-0.01354	-0.01341
	!EN	0.00654	0.00648	0.00641

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tinvt_1	!EN	0.01359	0.01364	0.01341
	!EN	-0.00632	-0.00648	-0.00641

Passive power(pJ) for EN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tinv_1	(A * !Y)	0.00000	0.37749	6.86663
	(A * !Y)	0.02866	0.40627	6.89527
	(!A * Y)	0.00809	0.38658	6.87217
	(!A * Y)	0.03222	0.41090	6.89651

Passive power(pJ) for EN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
gf180mcu_osu_sc_gp12t3v3__tinv_1	(A * !Y)	0.02145	0.40531	6.88906
	(A * !Y)	-0.00716	0.37672	6.86033
	(!A * Y)	0.02086	0.40042	6.88428
	(!A * Y)	-0.00346	0.37595	6.85994

GF180MCU_OSU_SC_GP12T3V3__XNOR2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__xnor2_1	52.92000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
gf180mcu_osu_sc_gp12t3v3__xnor2_1	0.00807	0.00800	7.98787

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__xnor2_1	0.00000	0.00284	0.00353

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__xnor2_1	A->Y (RR)	B	0.13675	2.90942	62.47430
	A->Y (FR)	!B	0.10057	4.74600	98.43540
	B->Y (RR)	A	0.11579	2.98765	64.92940
	B->Y (FR)	!A	0.11188	4.10452	86.63570

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__xnor2_1	A->Y (FF)	B	0.14346	3.05772	62.00250
	A->Y (RF)	!B	0.06981	2.80932	61.20540
	B->Y (FF)	A	0.11585	3.02223	61.95390
	B->Y (RF)	!A	0.08759	2.87542	61.80200

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__xnor2_1	A	B	0.02378	0.39457	6.93914
	A	B	0.05658	0.42712	6.96090
	A	!B	0.04976	0.55123	8.99276
	A	!B	0.00555	0.50664	8.94816
	B	A	0.00557	0.37744	6.90723
	B	A	0.04603	0.41753	6.94134
	B	!A	0.05896	0.58147	9.43520
	B	!A	0.00550	0.52789	9.37954

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__xnor2_1	A	B	0.05914	0.43047	6.95423
	A	B	0.02791	0.39965	6.92304
	A	!B	0.01463	0.50837	8.92055
	A	!B	0.05824	0.55212	8.96426
	B	A	0.05497	0.42732	6.94532
	B	A	0.01420	0.38698	6.90813
	B	!A	0.01534	0.52484	9.15520
	B	!A	0.06847	0.57791	9.21313

GF180MCU_OSU_SC_GP12T3V3__XOR2_1

gf180mcu_osu_sc_gp12t3v3_TT_25C.ccs
Cell Library: Process , Voltage 3.30,
Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Footprint

Cell Name	Area
gf180mcu_osu_sc_gp12t3v3__xor2_1	52.92000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
gf180mcu_osu_sc_gp12t3v3__xor2_1	0.00799	0.00802	7.96151

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
gf180mcu_osu_sc_gp12t3v3__xor2_1	0.00000	0.00284	0.00327

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__xor2_1	A->Y (RR)	!B	0.11593	2.98192	64.70250
	A->Y (FR)	B	0.11385	4.09988	86.42150
	B->Y (RR)	!A	0.14187	3.00540	64.72720
	B->Y (FR)	A	0.09640	4.03967	85.83090

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
gf180mcu_osu_sc_gp12t3v3__xor2_1	A->Y (FF)	!B	0.11579	3.01989	61.80080
	A->Y (RF)	B	0.08599	2.86979	61.63550
	B->Y (FF)	!A	0.12070	2.91467	59.14490
	B->Y (RF)	A	0.08553	3.50457	73.56620

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__xor2_1	A	B	0.06446	0.58728	9.44541
	A	B	0.01575	0.53833	9.39487
	A	!B	0.00416	0.37610	6.91341
	A	!B	0.04541	0.41691	6.94768
	B	A	0.05788	0.56736	9.24599
	B	A	0.01416	0.52377	9.20000
	B	!A	0.01344	0.38264	6.91995
	B	!A	0.04954	0.41865	6.94914

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
gf180mcu_osu_sc_gp12t3v3__xor2_1	A	B	0.00922	0.51860	9.15333
	A	B	0.05871	0.56798	9.20271
	A	!B	0.05624	0.42929	6.95424
	A	!B	0.01488	0.38870	6.91307
	B	A	0.00967	0.51240	8.87903
	B	A	0.05409	0.55855	8.92334
	B	!A	0.06114	0.43403	6.95863
	B	!A	0.02394	0.39706	6.92336